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THE IMPACT OF THE NON-INERTIAL BASE MOTION IN THE OPERATIONS OF ROBOTIC MANIPULATORS ON ORBIT

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Abstract

The robotic manipulators operation in orbit differs from those on ground. On ground the robotic manipulator base or platform is inertial in the sense that it does not move nor is disturbed by the motion of the robotic manipulator arms. In space the environment is characterized by microgravity and because of this the robotic manipulator base is not inertial. It moves and is disturbed by any motion of the robotic arm when performing operations of moving space components or astronauts from one place to other. This fact is associated with the Newton Law of Action and Reaction. The knowledge of the non-inertial base motion effects on the robotic manipulators operations is very important for the success of grasping objects in space, for berthing operations and manipulation of any device by using robotic manipulators. In particular the impact of non-inertial base will play a very important role for space tugs. This type of space vehicle shall be used in a near future to grasp, recover and repair on orbit spacecraft and to grasp objects such as debris. This article presents the mathematical model for a non-inertial robotic manipulator and analyzes the impact of the non-inertial base motion in the attitude control effort and performance. The equations of motion describe the translational and rotational motions. The translational motion describes the non inertial platform motion in response to the robotic arm displacements. The rotational motion describes the attitude motion. The dynamics analysis results are compared with a lab experiment with a robotic arm that operates on an ice non-inertial base to reduce the friction effect allowing the analyses of the platform reaction in response to the robot arm commanded actions. The robotic manipulator is commanded through wireless system by using Bluetooth.

Introduction

In 2007 the astronautic world commemorated the 50 years anniversary of the space age. It has been 50 years since a repetitive amazing beep-beep from orbit shook the old world order to its very foundations and started the space era. 1957 was the year the former Soviet Union launched the Sputnik. The first artificial satellite had nothing to do with the determination for the peaceful use of the outer space that guides the astronautical applications nowadays. It was a subject of ballistic-missile associated with the cold war, the environment of fear and aggression the USA and the former URSS got involved since the end of the World War II. Fifty years ago, if a country could launch a satellite into orbit it would at least raise a question of whether that country could use a missile to delivery an atomic bomb. Despite the fears and tension of the cold war days, the Sputnik marked the beginning of a golden age in the human understanding of the universe. The use of spaceflight for politics and propaganda purposes soon grew into much more as scientists began to realize the potential of such a profound change in perspective. Mankind's view of the universe, and of our blue planet in it, has changed significantly since the year of 1957. Remote sensing platforms have given scientists a extraordinary view of the planet Earth allowing comprehensive measurements of its working as a system of systems. On the other hand military reconnaissance satellites make it difficult to hide dangerous buildups and movements, giving potential military adversaries enough confidence to uphold treaties designed to let them pursue their competition by other means. That process has been hastened by satellite communications, which have shrunk the planet in ways that weaken despotic rule based on secrecy, restriction and fear. It is also true that hiding lives and values of different societies in the space age is much more difficult since satellite enable instantaneous voice, data and video transmissions from anywhere to anywhere. Since the sputnik hundred of space vehicles have been populating the space environment around the Earth and beyond. The outstanding progress of the space era has accomplished the ancient human dream to go to the Moon, to construct space stations, space telescope and other space vehicle, the sending of robots to Mars and automated vehicles beyond the solar system. Enormous quantity of new information and communication technologies has been growing up by side of the space era. New sensors and several electronic devices populate the new technologies related to space. New disciplines were born in the shadow of the challenges the scientists have faced to go through space. The man got no know and learns about working in the zero-g space environment. In this context the space robots, including the concept space telerobots[1] was born and has been growing up. The dreams of sending humans for the sake of space exploration come to a much more realistic and safety decision taking: to build and use robots to operate in space and other celestial bodies in replacement of the human being. The idea is consistent because of the fact that the space environment is hostile to living things in many ways. Sending humans to space for any kind of activity is very risky and highly expensive. Robots are possible to be replaced and improved thanks to the enormous advances of the information technology (IT) area. Operations that require robots are on-orbit rendezvous and docking/berthing (RVD/B) of spacecrafts[2-7], assembling space vehicle in space such as the International Space Stations, as have been used for the International Space Station (ISS), on-orbit servicing (OOS)[1] for maintaining and assistances to crew of ISS, scientific experiments in space and other planets. This is the direction of the future of the space exploration: use of intelligent robots and systems of robots with capability of local decision taking without the need to wait for commands and decisions from ground stations. The environment for the space operations is characterized by the apparent absence of gravity, known as microgravity environment, where everything experiences an absence of weight and fluctuates. In such environment the base or platform of any operation is non-inertial. The reaction forces associated with operations such docking, berthing, maintenance, grasping and moving other spacecraft or even space debris cause the non-inertial base or platform to react and move. Consider the following example as presented in[8]. Consider the remote manipulator system (RMS), mounted on a Shuttle of mass 67,000 Kg, and manipulating a 30,000 kg load. Commanding the RMS to move the load 6 m results a relative distance from the Shuttle of 6 m. However, due to opposite motion of the Shuttle of 1.8 m, it results in a net distance of only 4.2 m with respect to the target. So, the RMS miss its target by 1.8 m. The study of the response of the non-inertial platform to the robot motions is fundamental for the success of all the space operations to be taken in the microgravity orbital environment. The main goal of this paper is to study

the behavior of the non-inertial base or platform during OOS, mainly when the operation involves robot manipulators. The robot manipulators are widely used for extravehicular activities (EVA) by the astronauts. They are also used for experiment manipulations inside spacecraft. Other applications are in the process of berthing when a robot arm is used to grasp some place in the target spacecraft and in several concepts of the present and future space tugs. Another class of problem refers to the impact of the robot manipulators motion in the spacecraft attitude and accelerations disturbances to the satellite. These disturbances may affect on-board experiments, mainly those requiring fine pointing of instruments. Control efforts to maintain and deal with the nominal attitude increases when the manipulators are working. Planed target trajectory to reach and capture an object or to berth with another spacecraft may fail if the non-inertial platform is free to respond to the robot effort in their on-orbit operations. A comprehensive study is required to prevent control failures and to establish the correct requirement specifications of the hardware such as actuators and sensors that composes the attitude and orbit control subsystem (AOCS) for the space mission. This paper involves two types of studies. One is with respect to modeling and simulation of the equations of the dynamics for the sake of attitude control analysis during on-orbit operations with a spacecraft containing robotic manipulators. The other is with respect to experiment in laboratory to gain more insight about the non-inertial reactions of the manipulator base or platform during operations. We cannot reproduce the space environment with respect to microgravity for a reasonable time to analyze the attitude and control operations. However, it is possible to represent the non-inertial fixed base by using an air-supported platform or even ice platform on a glass table. For this paper the results were obtained by using ice platforms to approach the non-inertial fixed base. The lab experiments confirmed the platform reaction motion due to the robot manipulator operations. The simulation results for attitude have shown that the control efforts to keep the platform stabilized are greater than that efforts to maintain a prescribed attitude when the robot manipulator is not in operation.

Mathematical Modeling

We have developed two set of equations of motions. One set of equations was developed by considering the platform three rotational degrees-of-freedom for the rotational motion plus two DOF for the robotic manipulator. This mathematical model was for the simulation of motion via computer by using the MatLab® software (SW) package. The simulations aims the attitude control and control effort evaluation for both the cases, one in which the robotic arm is not in operation and the other when the manipulator is moving on the platform. the glass table in the Mechatronics lab. The mathematical model is a simplification of the first set of equation by restricting the motion on the surface of the glass table. Two degrees of freedom are taken off of the first set of equation, that is, the rotational motion is restricted to a pure rotation about the normal to the glass table.. Figure 1 illustrates the physical model for first set of equations given by Eqs.(1) to Eq.(5). The equations of motion have been derived by the Lagrangian formulation for quasi-coordinates and the Lagrangian formulation for generalized coordinates[8] Figure 1 illustrates the physical model for the chaser spacecraft attitude motion

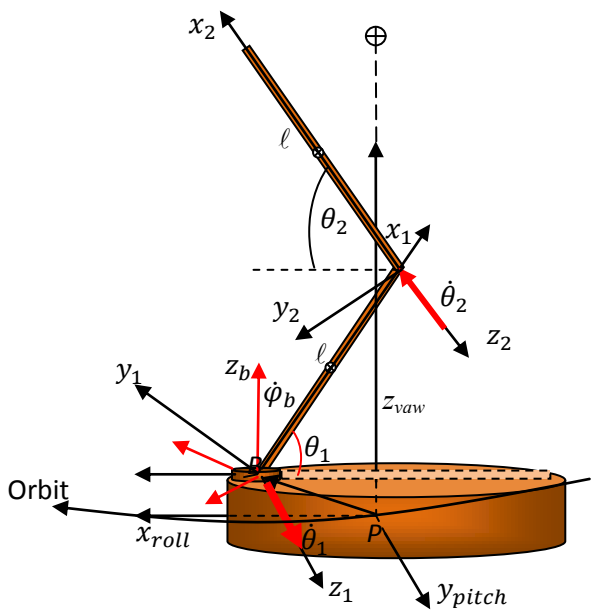

Fig. 1 - Chaser Spacecraft physical model

The Euler Modified Equations for the model shown in figure 1 can be written as

$$J_{t_x}\dot{\omega}_{x_t} - I_{xz_{12}}\dot{\omega}_z + (J_{z_t} - J_{y_t})\omega_y(\omega_z + \dot{\varphi}) + [I_{y_1}s(2\theta_1)\dot{\theta}_1 + I_{y_2}s(2\theta_2)\dot{\theta}_2]\omega_x + \{[[2I_{y_2}c(2\theta_2) + I_{z_2}](\dot{\theta}_2 - \dot{\theta}_1) - 2I_{y_1}c(2\theta_1) + I_{z_1}]\dot{\theta}_1\}(\omega_z + \dot{\varphi}) - I_{xz_{12}}\omega_x\omega_y = \tau_x + \tau_{m_{12}} - u_\varphi \quad (1)$$

$$J_{y_t}\dot{\omega}_y + +(J_{z_t} - J_{x_t})\omega_x(\omega_z + \dot{\varphi}) + I_{xz_1}((\omega_z + \dot{\varphi})^2 - \omega_x{}^2) = \tau_x + u_{\theta_1} + u_{\theta_2} \quad (2)$$

$$J_{x_t}\dot{\omega}_z - I_{xz_{12}}\dot{\omega}_x + (J_{y_t} - J_{x_t})\omega_x\omega_y - [I_{y_1}s(2\theta_1)\dot{\theta}_1 + I_{y_2}s(2\theta_2)\dot{\theta}_2]\omega_x + \{[[2I_{y_2}c(2\theta_2) + I_{z_2}](\dot{\theta}_2 - \dot{\theta}_1) - 2I_{y_1}c(2\theta_1) + I_{z_1}]\dot{\theta}_1\}(\omega_z + \dot{\varphi}) - I_{xz_{12}}\omega_y(\omega_z + \dot{\varphi}) = \tau_x - \tau_{xz_{12}+}u_\varphi \quad (3)$$

$$I_{z_{12}}(\ddot{\theta}_1 + \dot{\omega}_y) + I_{z2}(\ddot{\theta}_2 - \ddot{\theta}_1) + [I_{y1}s2\theta_1 + I_{y2}s2\theta_2](\omega_z^2 - \omega_x^2) + 2[I_{y1}c2\theta_1 + I_{y2}c2\theta_2]\omega_x\omega_z = u_{\theta_1} \quad (4)$$

$$I_{z_2}(\ddot{\theta}_2 - \ddot{\theta}_1 + \dot{\omega}_y) + I_{y_2}s2\theta_2(\omega_z^2 - \omega_x^2) + 2I_{y2}c2\theta_2\omega_x\omega_z = u_{\theta_2} \quad (5)$$

where

$\dot{\varphi}$ is the relative angular rate of the robotic manipulator with respect to the platform z axis.

θ_1 is the angular time position of the first robotic manipulator arm

θ_2 is the angular time position of the second robotic manipulator arm.

ω_x , ω_x , and ω_z are the angular rates of the complete system (the space vehicle containing the robot manipulator) about the platform *x, y,* and *z* axes, respectively

$u = I_{z_m}\ddot{\varphi}$, torque actuating on the first joint of the manipulator

$u = I_{xz_{12}}(\ddot{\theta}_1)$, torque actuating on the second joint of the manipulator

$u = I_{z_{12}}(\ddot{\theta}_2)$, torque actuating on the third joint of the manipulator

The inertia properties and symbols are:

J_{x_t}, J_{y_t} , and J_{z_t} are the total (t) principal moments of inertia of the space vehicle

I_{z_1} and I_{z_2} are the moments of inertia of the arm 1 and 2 of the robotic manipulator, respectively. For this study $I_{x_1} = I_{x2} = 0$ and $I_{z_1} = I_{y_1}, I_{z_2} = I_{y_2}$.

u_α is the torque associated with the rotational DOF of the robotic manipulator, $\alpha = \theta_i$ $(i = 1,2)$ and φ.

The second set of equations is for the lab tests. For on-orbit motion there are considered three DOF for attitude while for the lab the attitude is described by ne DOF, on the plane of the glass table. Figure 2 illustrates the physical model for the laboratory tests.

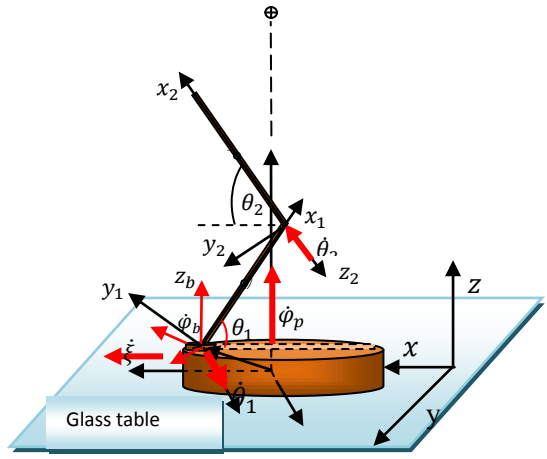

Fig. 2 – Physical model for the lab tests

The equations of motion for this model are

$$J_{x_t}\dot{\omega}_z + \{[[2I_{y_2}c(2\theta_2) + I_{z_2}](\dot{\theta}_2 - \dot{\theta}_1) - 2I_{y_1}c(2\theta_1) + I_{z_1}]\dot{\theta}_1\}(\omega_z + \dot{\varphi}) = \tau_x - \tau_{xz_{12}+}u_\varphi \quad (6)$$

$$I_{z_{12}}(\ddot{\theta}_1) + I_{z2}(\ddot{\theta}_2 - \ddot{\theta}_1) + [I_{y1}s2\theta_1 + I_{y2}s2\theta_2](\omega_z^2) = u_{\theta_1} \quad (7)$$

$$I_{z_2}(\ddot{\theta}_2 - \ddot{\theta}_1 + \dot{\omega}_y) + I_{y_2}s2\theta_2(\omega_z^2) = u_{\theta_2 -}u_{\theta_1} \quad (8)$$

$$I_{zm}\ddot{\varphi} = \tau_\varphi \quad (9)$$

The physical model for the relative translational motion between the center of mass of the chaser and target spacecraft is illustrated in figure 3.

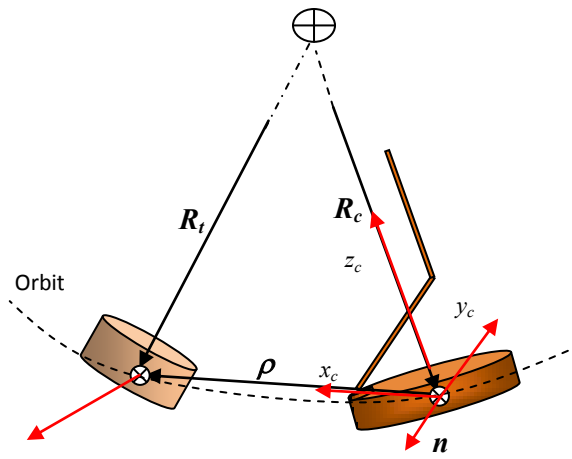

Fig.3 - Physical model for the translational motion between the chaser and target spacecraft.

In figure 3, $\boldsymbol{R}$ is vector from the center of the Earth to the center of mass of the chaser and target spacecrafts. The subscripts t and c means target and chaser, respectively. The difference between $\boldsymbol{R_t}$ and $\boldsymbol{R_c}$ is $\boldsymbol{\rho}$ .The Hill's equation of motion for the relative motion can be written as:

$$\ddot{x} - 2n\dot{z} = f_x \quad (10)$$

$$\ddot{y} + n^2 y = f_y \quad (11)$$

$$\ddot{x} + 2n\dot{x} - 3n^2 z = f_z \quad (12)$$

This formulation with the analytical solution for the homogeneous system is shown in Bong We[9].

Lab experiments

The experiment is conducted in the Mechatronics Laboratory of the Dpt. Of Mechatronics, Mechanical and Aeronautics Engineering Division, ITA. The planned experiment infrastructure consists of glass table, two floating robotic manipulators (see figure 4 for the chaser manipulator), an intelligent sensor (Kinect) for tracking the manipulators motions on the table and a communication subsystem to receive data from the sensor and issue commands to the robotic. Figure 5 shows the manipulator hardware communication configuration. This manipulator is an Open Development as an Open Hardware project[10]. The user sends the commands to the arduino by using a computer equipped with the Bluetooth technology. The four servos receive the user commands through the interface with one of the arduinos. The arduinos are connected by wire while the Bluetooth is a communication wireless technology. Figure 6 shows the degrees-of-freedom of the robotic manipulator .The manipulator can rotates o its own base to better performance in its work space. There three DOF related to manipulator arms and at the tip of the second link, the grasping device.

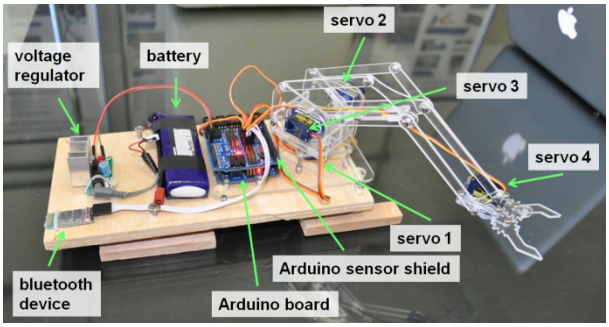

Fig. 4 – The chaser robotic manipulator

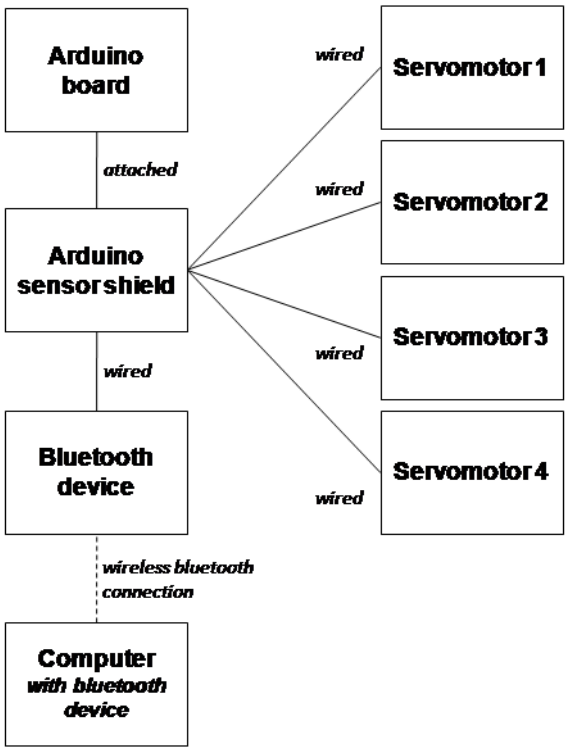

Fig. 5 Robotic Manipulator Hardware Communication Configuration

This communication subsystem consists of a laptop and the Bluetooth technology to issue commands to the manipulator. This system is human operated and put the complete system in the category of telerobots[1]. This system is not read yet for the planned experiment.

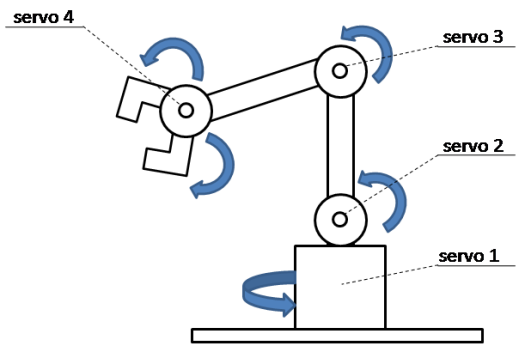

Fig. 6 – DOF of the robotic manipulator

For this paper it was used a prototype constituted by one manipulator and the communication system as described above. The measurements of the reactive linear effects of the manipulator was computed by using a camera to record the motion and then treated later to obtain the reaction effects of the manipulator motion on the non-inertial platform. The same procedure was used for the computation of the rotational effect of the platform. This platform has been planned to use a air compressed system so as to allows the robot to float on a thin shell of air. However, in this very first prototype we have used ice platforms to represent the non-inertial base as proposed in the scope of this paper. The innovation in the experiment is the usage o the Bluetooth and the Kinect technologies. The Kinect is a device of sensing movements developed by Microsoft for the console Xbox 360. It is equipped with a cameras system that allows the user to control and interacts with the console without having to touch a game controller, but just using gestures or voice commands. This device is powered by both hardware and software that execute fundamental tasks for this project: it generates a three-dimensional image of objects in its field of view and recognizes human beings amount them. Furthermore, the Kinect (Figure 7) have also a stereo microphone to allow conversations and voice commands. It is provided of an infrared light source, 3D depth camera and a RGB camera.

Fig. 7 - Kinect Sensor

Early software used to use the technique of colors and textures differentiation to distinguish objects from their backgrounds. The Kinect uses a different technique. The infrared light source transmits light and the sensor measures the time of travel after these waves reflect in the objects. The time of travel works likes the sonar: if someone knows how long the light takes to return, then it is possible to know how far away one object is. The infrared generator partially solves the problem of ambient light, since the sensor is not designed to record visible light. As information is returned, some of it is deformed - which can help to generate an image of finer texture of 3-D objects and not just its depth. Using this technique the Kinect can differentiate depths of objects around 1 cm and 3 mm in length and width. Kinect sensor has been studied by the scientific community as a promising innovative sensor. These studies and publications intensified after the Microsoft release of the .NET Kinect interface in early 2011.

Simulations and test Results

The Eq.(1) to Eq.(5) linear associated equations have been integrated via MatLab® by using the LQR optimal control to check the control effort for controlling the attitude motion for the robot spacecraft by keeping the robot manipulator out of work and by considering the robotic manipulator working. The expectation was to have less effort when controlling the spacecraft with manipulator out of work and more effort control to stabilize the attitude when the robot manipulator is working. The results confirm that the control effort differs for both spacecraft configurations. The matrix coefficient of the pitch, roll, and yaw accelerations are not constant but dependents of the robotic manipulator positions. As consequence of the deployment of the robot arms, perturbations are induced on the dynamics and the more control effort is required to keep the nominal attitude during the close proximity of the chase and target spacecrafts. Figure 8 to 13 shows the results.

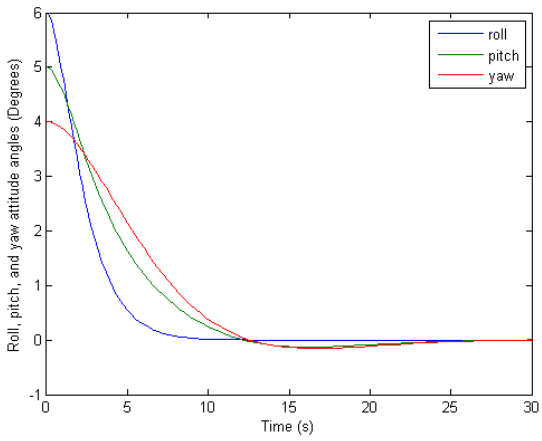

Fig. 8 – Attitude control action to reach the prescribed attitude, arms retracted

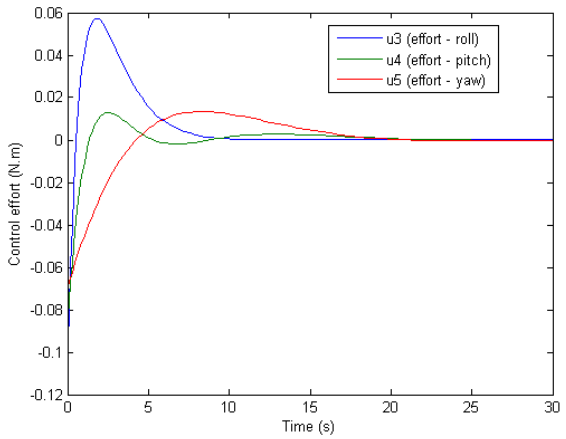

Fig. 9 Control efforts to control the spacecraft attitude, manipulator arms retracted

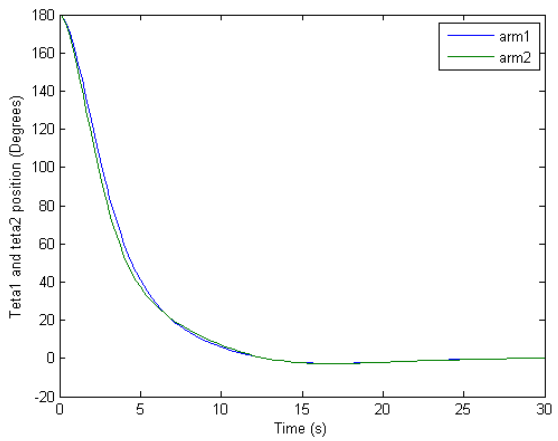

Fig. 10 – Arms being retracted to the zero position

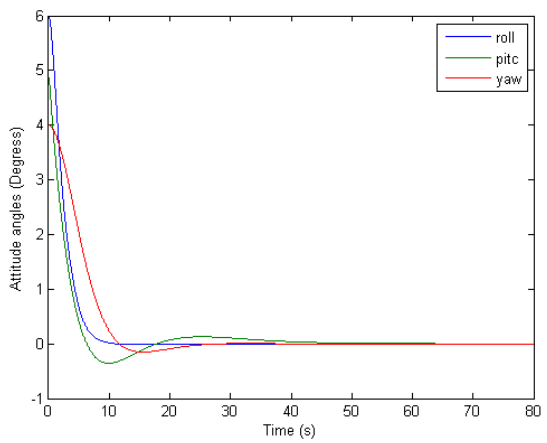

Fig. 11 Attitude control while the arms are being retracted

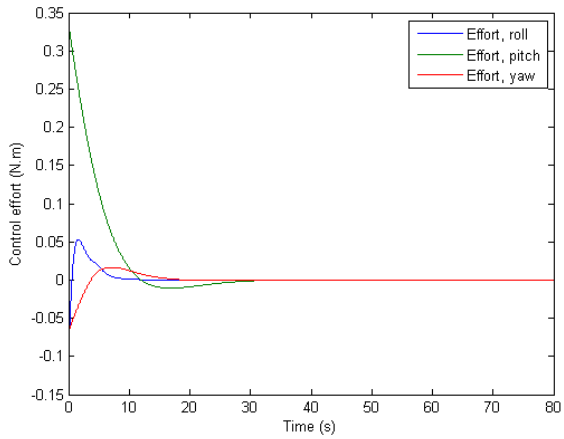

Fig. 12 - Attitude Control Effort during the arms retraction

The Null Reaction Case

It is possible to control the reaction on the platform by using actuators. For attitude one of the actuators is reaction wheels. It absorbs the excitation caused by the rotation of the robotic manipulator. It is required to prevent the wheels saturation by a de-saturation procedure. For robotic manipulators there are design options to eliminate the reactions torques. The physical models shown in figures 1 and 2, the arms deploys in opposite sense and then the reaction on the base is null if the arms are deployed in a synchronous mode. Figure 13 shows the results for such configuration. To simulate this case the attitude is first taken to the nominal attitude, here assumed as zero degrees in roll, pitch and yaw. Then the arms are deployed without any additional control effort to keep the attitude close to zero.

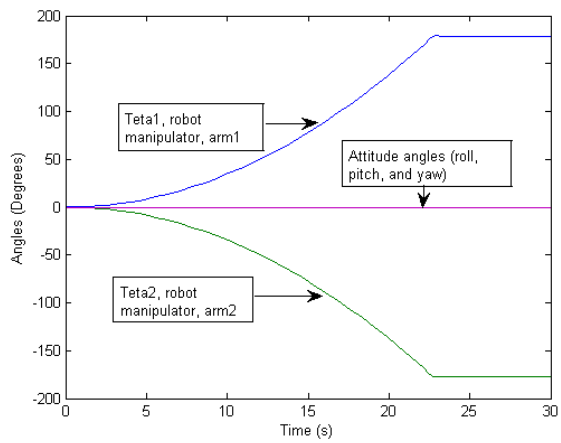

Fig. 13 – The arms are deployed - null reaction configuration case

Experiments

The experiments in the lab aimed to analyze the impact of the robotic arms on the platform rotational and translational motion. Ice platforms have been used to represent the non-inertial base on a glass table. A camera has been used to record the motion and then the video images have been treated by using the software Kinovea[11]. Figure 14 shows one of the video images. From the video images processing, the ice non-inertial base displacements and rotation angles were obtained and are shown in table 1 and 2, respectively.

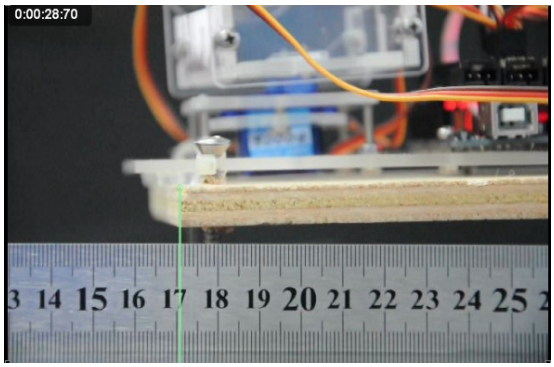

Fig. 14 – Video images processing to obtain the non-inertial base displacement and angles.

Figure 15 illustrates the non-inertial base response to the robot arm motions based on the data shown in table 1.

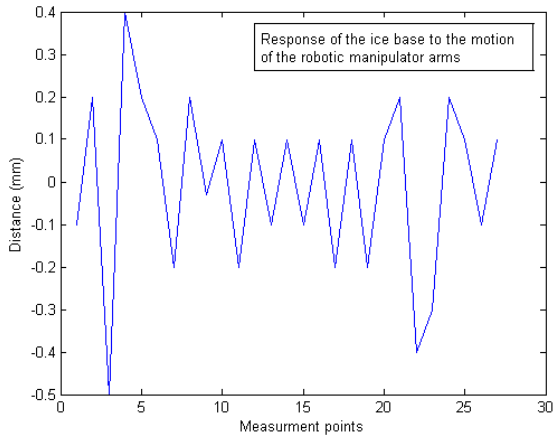

Fig. 15 – Ice Non-inertial base response to the robot manipulator arms motion.

Table 1: motion of the non-inertial platform (ice platform) in response to the robotic manipulator motion

						d1		d2	
P1	**d1**	**P2**	**d2**	**P3**	**motion**	shrink	stretch	shrink	stretch
18,2	-0,1	18,1	0,3	18,4	shrink	-0,1		0,3	
18,4	0,2	18,6	-0,2	18,4	stretch		0,2		-0,2
18,4	-0,5	17,9	0,2	18,1	shrink	-0,5		0,2	
18,1	-0,4	17,7	0,0	17,7	stretch		-0,4		0
17,7	0,2	17,9	-0,1	17,8	shrink	0,2		-0,1	
17,8	0,1	17,9	-0,3	17,6	stretch		0,1		-0,3
17,6	-0,2	17,4	0,0	17,4	shrink	-0,2		0	
17,4	0,2	17,6	-0,3	17,3	stretch		0,2		-0,3
17,3	-0,3	17,0	0,3	17,3	shrink	-0,3		0,3	
17,3	0,1	17,4	-0,4	17,0	stretch		0,1		-0,4
17,1	-0,2	16,9	-0,2	16,7	shrink	-0,2		-0,2	
16,7	0,1	16,8	-0,3	16,5	stretch		0,1		-0,3
16,5	-0,1	16,4	0,1	16,5	shrink	-0,1		0,1	
16,5	0,1	16,6	-0,3	16,3	stretch		0,1		-0,3
16,3	-0,1	16,2	0,1	16,3	shrink	-0,1		0,1	
16,2	0,1	16,3	-0,4	15,9	stretch		0,1		-0,4
15,9	-0,2	15,7	0,4	16,1	shrink	-0,2		0,4	
16,1	0,1	16,2	-0,4	15,8	stretch		0,1		-0,4
15,8	-0,2	15,6	0,1	15,7	shrink	-0,2		0,1	
15,7	0,1	15,8	-0,4	15,4	stretch		0,1		-0,4
15,4	0,2	15,6	-0,1	15,5	shrink	0,2		-0,1	
15,5	-0,4	15,1	0,3	15,4	stretch		-0,4		0,3
15,1	-0,3	14,8	0,3	15,1	shrink	-0,3		0,3	
15,1	0,2	15,3	-0,5	14,8	stretch		0,2		-0,5
14,8	0,2	15,0	0,0	15,0	shrink	0,2		0	
15,0	0,1	15,1	-0,4	14,7	stretch		0,1		-0,4
14,7	-0,1	14,6	0,1	14,7	shrink	-0,1		0,1	
14,6	0,1	14,7	-0,4	14,3	stretch		0,1		-0,4
14,3	0,2	14,5	0,0	14,5	shrink	0,2		0	
average	0,0		-0,1			-0,1	0,1	0,1	-0,3

Table 2: Rotational response of the non-inertial base to the robotic manipulator rotation

A1	d1	A2	d2	A3	d3	A4	d4	mov
14º	12º	26º	-9º	17º	-13º	4º	10º	L→R
14º	14º	28º	-8º	20º	-12º	8º	15º	R→L
23º	14º	37º	-3º	34º	-13º	21º	15º	L→R
36º	16º	52º	-9º	43º	-14º	29º	9º	R→L
38º	17º	55º	-8º	47º	-18º	29º	17º	L→R
46º	21º	67º	-11º	56º	-20º	36º	9º	R→L
45º	17º	62º	-9º	53º	-13º	40º	12º	L→R
52º								
average	**15,9º**		**-8,1º**		**-14,7º**		**12,4º**	

In table 1, P1 is the initial position of the ice platform, P2 is the position where the position is reverted, P3 is the end position, d1 is the distance between P' and P3, d2 is the distance between p2 and p3 and d3 is the distance between P3 and P1 of the next motion. In table 2 A1 is the initial angle, A2 is a point of reversion of the ice base rotation in response to the reversion of the arm rotation to its initial position, A3 is the final angle. D1 = A2-A1; D2 = A3-A2, and D4=A4-A1.

Conclusion

This work deals with the dynamics analysis of robotics manipulators on non-inertial platforms aiming space applications. Three sets of equations of motion have been derived by using the Lagrangian formulation for quasi coordinates to obtain the modified Euler Equations of motion and the Lagrangian formulation for generalized coordinates to obtain the robotic manipulator equations of dynamics. This mathematical model has been used to simulate the orbital operations in the microgravity environment where the bodies experience floating situations and the platforms become a non-inertial fixed base. Through simulations it was verified that the control effort increases when the space vehicle implement the attitude control to take the spacecraft to a three-axis prescribed attitude in roll, pitch, and yaw. The second set of equations was derived to compare the simulations results with the lab experiments. The third set of equations was derived for the sake of analysis of the close proximity phase between the chaser and the target vehicle during the rendezvous and docking/berthing operations. Also the study of these dynamics configurations is checked against the experiments in laboratory where ice platforms have been used to simulate the orbital motion. It were checked the reactive motions of the non-inertial base when the robotic manipulator is commanded for grasping the target and when the arms are retracted to change the mode operation. There are two operation modes for the robotic manipulator. One is the manipulator rotation to find the right direction towards the target. The second mode is the stretching of the arms to acquire the target.

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